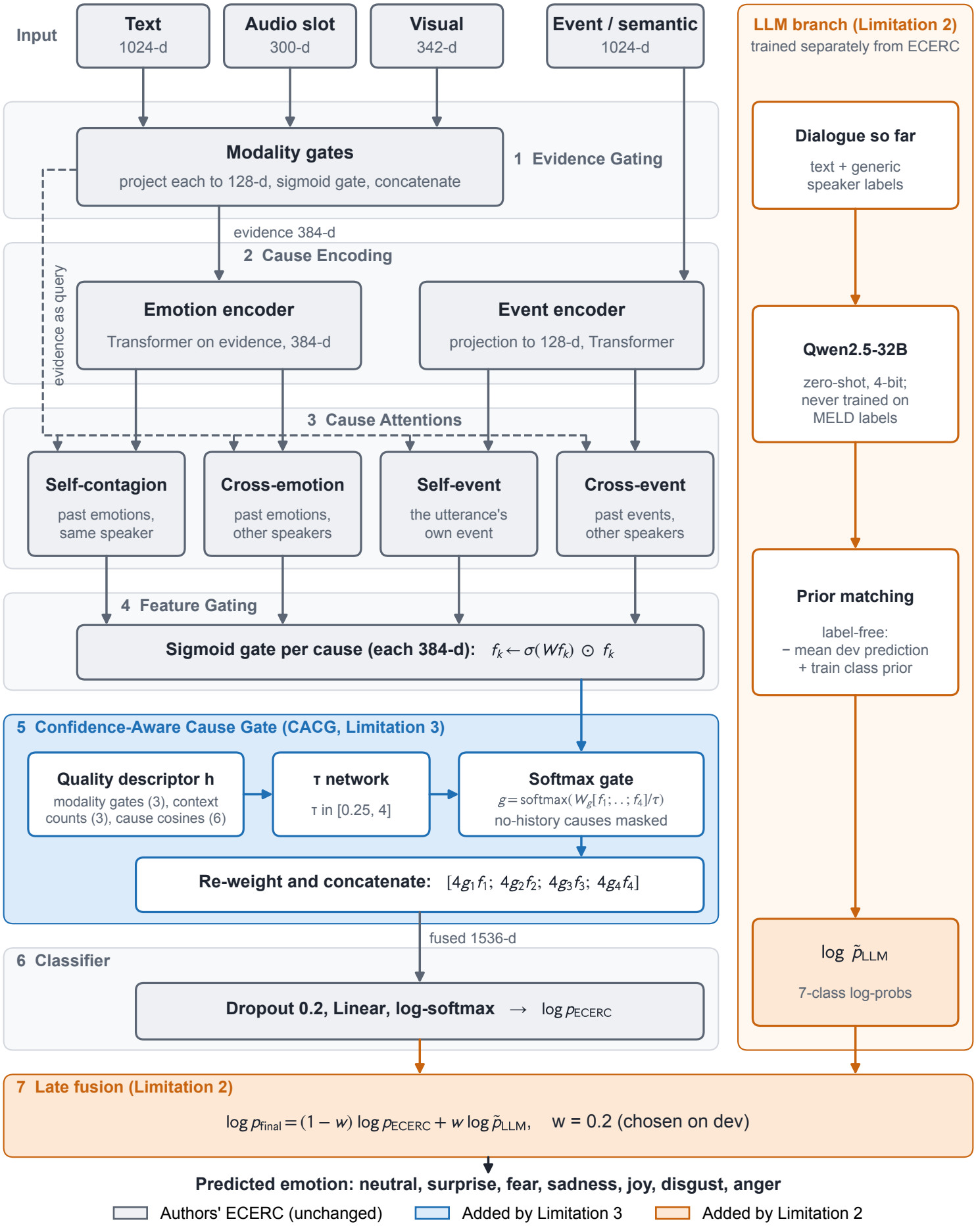

Input
Text
1024-d
Audio slot
300-d
Visual
342-d
Event / semantic
1024-d
Modality gates
project each to 128-d, sigmoid gate, concatenate
1 Evidence Gating
evidence 384-d
2 Cause Encoding
evidence as query
Emotion encoder
Transformer on evidence, 384-d
Event encoder
projection to 128-d, Transformer
3 Cause Attentions
Self-contagion
past emotions, same speaker
Cross-emotion
past emotions, other speakers
Self-event
the utterance's own event
Cross-event
past events, other speakers
4 Feature Gating
Sigmoid gate per cause (each 384-d): $f_k \leftarrow \sigma(Wf_k) \odot f_k$
5 Confidence-Aware Cause Gate (CACG, Limitation 3)
Quality descriptor h
modality gates (3), context counts (3), cause cosines (6)
τ network
τ in [0.25, 4]
Softmax gate
$g = \mathrm{softmax}(W_g[f_1; \ldots; f_4]/\tau)$
no-history causes masked
Re-weight and concatenate: $[4g_1f_1;\ 4g_2f_2;\ 4g_3f_3;\ 4g_4f_4]$
fused 1536-d
6 Classifier
Dropout 0.2, Linear, log-softmax → $\log p_{\mathrm{ECERC}}$
LLM branch (Limitation 2)
trained separately from ECERC
Dialogue so far
text + generic speaker labels
Qwen2.5-32B
zero-shot, 4-bit; never trained on MELD labels
Prior matching
label-free:
− mean dev prediction
+ train class prior
$\log\ \tilde{p}_{\mathrm{LLM}}$
7-class log-probs
7 Late fusion (Limitation 2)
$\log p_{\mathrm{final}} = (1-w)\log p_{\mathrm{ECERC}} + w \log \tilde{p}_{\mathrm{LLM}}$, w = 0.2 (chosen on dev)
Predicted emotion: neutral, surprise, fear, sadness, joy, disgust, anger
Authors' ECERC (unchanged)
Added by Limitation 3
Added by Limitation 2